

PPC-WIM - CBE PPC Inventory Traceability System

MVP Document

Project 26012 | Prepared by Pinesphere Team | 12 Sep 2026

1. Introduction

Watch cases sent to vendors for polishing and plating come back to Titan CBE, where the PPC and Stores teams verify them against the delivery challan, count the pieces and look for rack space by hand.

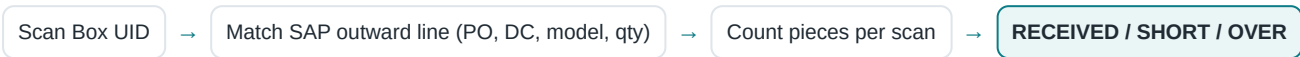
PPC-WIM digitises this inward flow. Each dispatched lot gets a Box UID, the returned box is scanned to verify it against its SAP outward line, and its exact rack, shelf, row and tray position is recorded so it can be found later without searching.

2. DFD and Workflow - Process 6, 17, 18

The inward DFD has 19 processes. The MVP covers the three highlighted below.



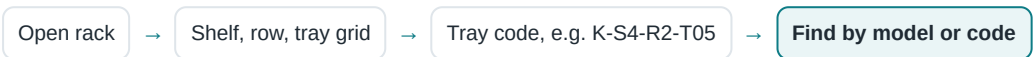
Process 6 - Verify received box (SAP Inward screen)



Process 17 - Move to CBFC rack (Rack Locator)

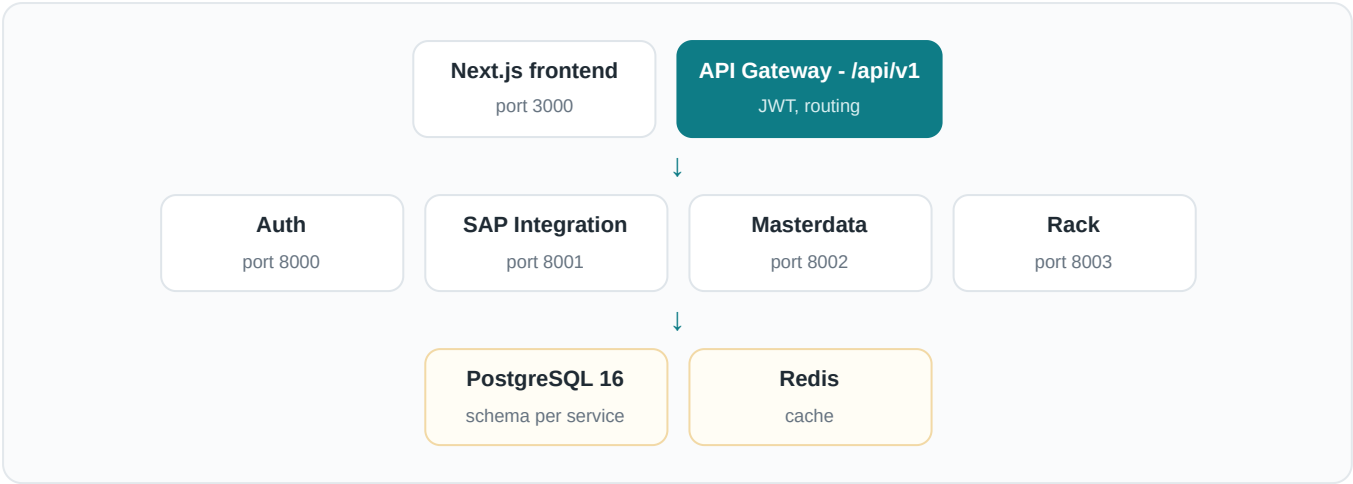


Process 18 - Rack, Row, Bin mapping (Rack Locator)



3. Technical Architecture

The frontend talks to one API gateway. Behind it, each service runs on its own port and owns its own database schema.

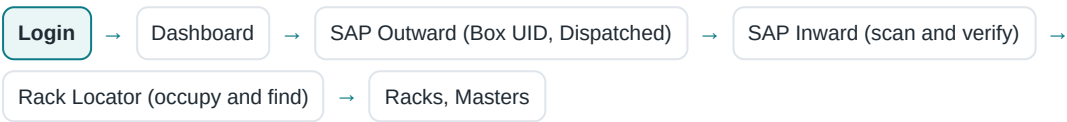


Process	Service	API	Data
P6 - Verify	Masterdata	POST /sap-inward/scan, /close, /reset	sap_outwards tally, sap_inward_scans
P17 - Move	Rack	GET /rack/locate, POST /slots/{id}/occupy	rack tray slot: state, model, lot, qty
P18 - Map	Rack	GET /rack/racks/{code}, /resolve, /find	rack_master and materialised tray slots

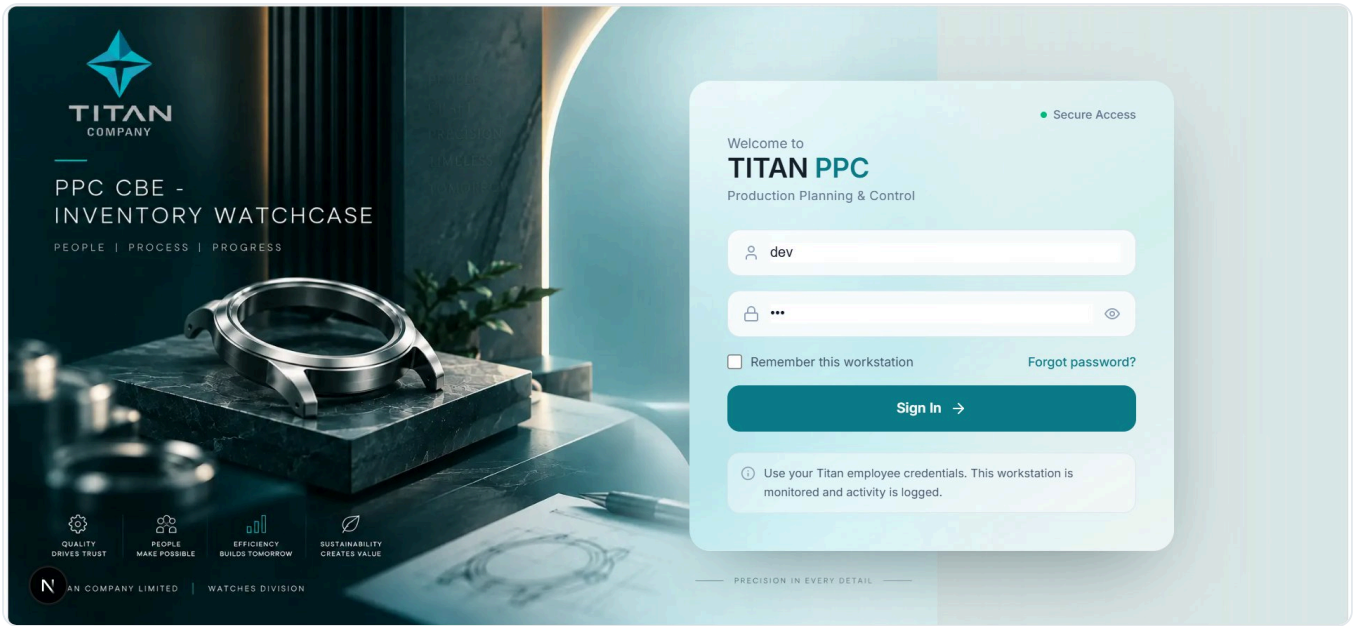
4. Tech Stack

Frontend	Next.js 15, React 19, TypeScript, Tailwind CSS v4, TanStack Query
Backend	Python 3.12, FastAPI, SQLAlchemy 2.0, Alembic
Database	PostgreSQL 16, Redis 7
Security	JWT with refresh cookie, role-based access
Deployment	Docker Compose, Nginx, on-premise

5. Application Flow



6. User Interface



Login

The operator signs in with a Titan employee ID and password, and a JWT session is issued for every screen behind it. After sign-in the user lands on the Dashboard and works through SAP Outward, SAP Inward and Rack Locator.

TITAN PPC

Search DC, PO, material, vendor, batch...

Scan

LIVE

10:56:56
11 Sept 2026

Developer
Owner

SAP Outward > Outward Records

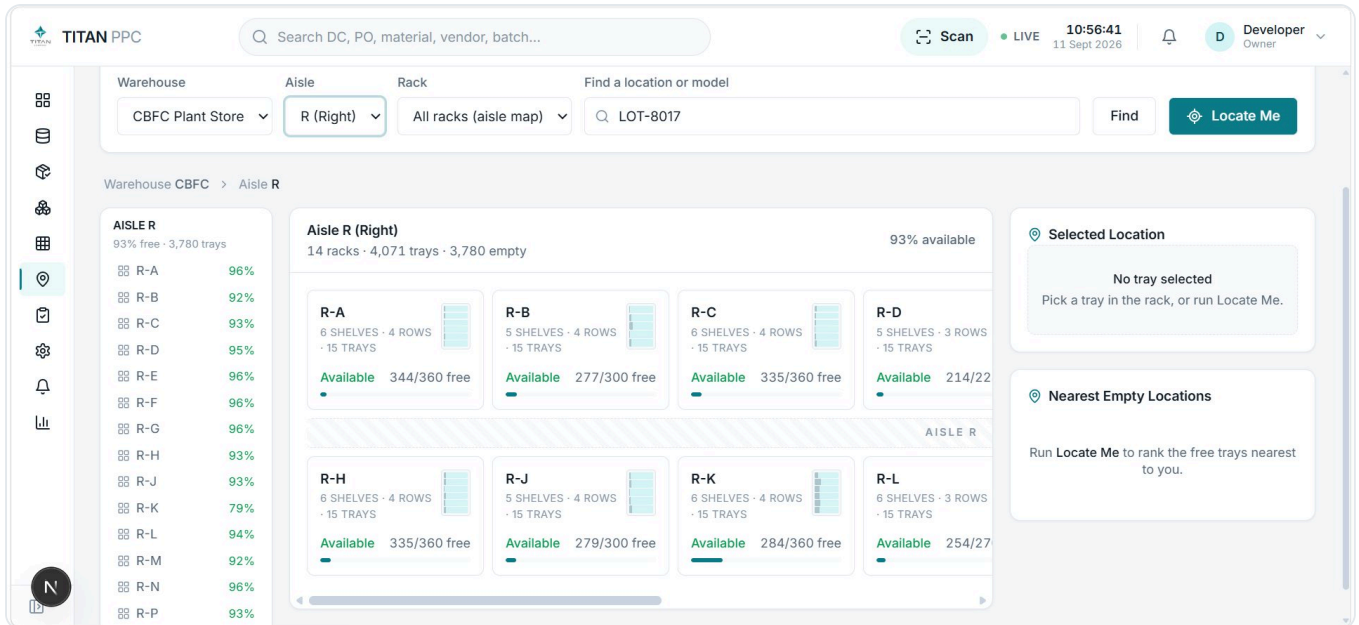
SAP Outward

Search DC, PO, model, vendor, batch...

S.No	Date / Timestamp	DC no	PO no	Model	Vendor	Batch no	SAP	Lot Qty	Tray Type	No. of Trays	Front Case Trays	Back
1	10/9/2026, 10:30:00 pm	DC-260910-05	PO-4500002522	90142	Kalai Industries	B260904	101	467	FC + BC	38	19	
2	10/9/2026, 10:30:00 pm	DC-260910-35	PO-4500006946	90148	Kalai Industries	B260933	101	263	FC + BC	22	11	
3	10/9/2026, 10:30:00 pm	DC-260910-05	PO-4500008446	90127	Shine Times	B260912	101	206	FC + BC	18	9	
4	10/9/2026, 10:30:00 pm	DC-260910-22	PO-4500006672	90148	Shine Times	B260924	101	347	FC + BC	28	14	
5	10/9/2026, 10:30:00 pm	DC-260910-05	PO-4500007476	90127	Shine Times	B260968	101	389	FC + BC	32	16	
6	10/9/2026, 10:30:00 pm	DC-260910-18	PO-4500002889	90142	Kalai Industries	B260994	101	394	FC + BC	32	16	
7	10/9/2026, 10:30:00 pm	DC-260910-01	PO-4500005545	90102	Shine Times	B260958	101	210	FC + BC	18	9	
8	10/9/2026, 10:30:00 pm	DC-260910-19	PO-4500000067	90148	Shine Times	B260952	101	274	FC + BC	22	11	

SAP Outward

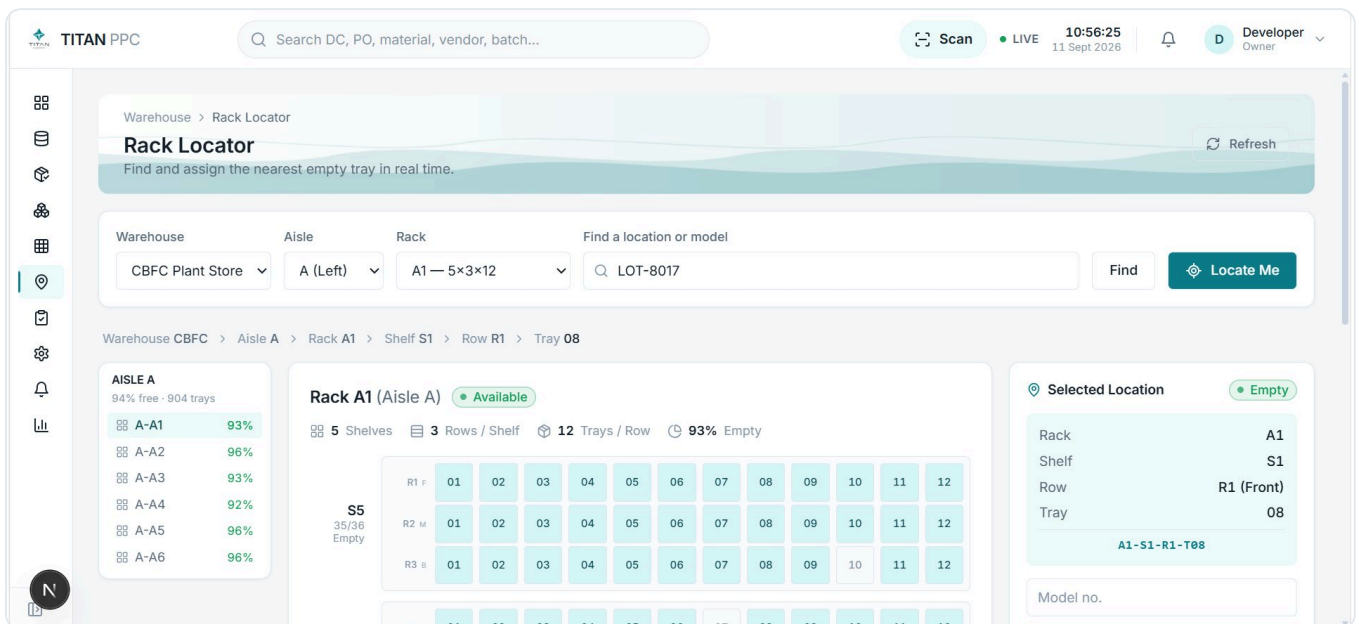
SAP outward lines are synced in with DC, PO, model, vendor, batch, SAP movement (101) and lot quantity. The operator fills in tray type, number of trays and front / back case trays, then assigns the Box UID. The line becomes Dispatched and is verified again at inward (Process 6).



Rack Locator - aisle map

Pick a warehouse and aisle to see every rack with its shelf x row x tray shape and the live number of free trays (for example R-K 284/360 free).

Locate Me ranks the nearest empty trays, so the returned box can be moved to a CBFC rack without searching (Process 17).



Rack Locator - rack detail

Open a rack to see its shelf, row and tray grid, then click a tray or type a code such as A1-S1-R1-T08 to select it.

Enter the model number to occupy the tray. The exact rack, shelf, row and tray is stored and can be searched by model or lot (Process 18).